Dynamic Guideline: WHO-Based Poisoning - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope clarification

A targeted WHO search identified **five WHO poisoning-related guideline/technical guidance documents** suitable for synthesis. The most recent WHO poisoning-specific guideline identified is **"Guidelines for establishing a poison centre," published 14 January 2021**, which WHO describes as an update of the 1997 WHO/IPCS *Guidelines for poison control* WHO poisoning page, WHO departmental update.

The requested title pattern **"Meaningful youth engagement in a WHO guideline development process: case study of WHO [] guideline"** was found as **"Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)," dated 11 December 2025**. It is **not a poisoning guideline** and is therefore not counted among the five poisoning guidance documents, but its methods are relevant to future inclusive poisoning-guideline development WHO publication page, WHO webinar page.

Exactly five WHO poisoning-related guidelines/technical guidance documents synthesized

No.	WHO guideline / technical document	Publication date / status identified	Target population / users	Scope and objectives
1	**Guidelines for establishing a poison centre**	14 January 2021	Ministries of health, poison-centre planners,	Establish and operate poison centres; develop poison-information, clinical toxicology, analytical

No.	WHO guideline / technical document	Publication date / status identified	Target population / users	Scope and objectives
			clinicians, public-health authorities	toxicology, toxicovigilance, antidote-access, training, quality, funding and chemical-incident response functions WHO departmental update, WHO poisoning page.
2	**Guidelines for poison control**	10 June 1997	National poison-control programmes, poison information services, clinical toxicology services	WHO/IPCS guidance on poison-control systems, poison information centres, poisoning prevention and management; updated by the 2021 guideline WHO publication page, PAHO/WHO.
3	**WHO Recommended Classification of Pesticides by Hazard and guidelines to classification, 2019 edition**	1 May 2020	Regulators, public-health authorities, agriculture and chemical-safety sectors	Classification of pesticides by hazard to support risk management and prevention of pesticide poisoning WHO poisoning page.
4	**Clinical management of acute pesticide intoxication: prevention of suicidal behaviours**	WHO IRIS technical document identified	Clinicians, emergency-care systems, mental-health and suicide-prevention programmes	Clinical management of pesticide intoxication and linkage to suicide-prevention responses WHO IRIS PDF.
5	**IPCS INTOX Poisons Centre**	WHO/IPCS training	Poison-centre staff, clinicians,	Practical training on poison-call handling,

No.	WHO guideline / technical document	Publication date / status identified	Target population / users	Scope and objectives
	Training Manual	guidance identified	toxicology trainees	emergency stabilization, documentation, exposure history, toxicovigilance and general management of poisoned patients Trainer manual, Trainee manual.

Public-health burden and rationale

WHO describes poisoning from **pharmaceuticals, industrial chemicals, pesticides, chemical products and natural toxins** as a significant global public-health problem WHO poisoning page. WHO-reported indicators include:

- **106,683 deaths from unintentional poisoning in 2016** WHO departmental update.
- **20% of global suicides due to pesticide self-poisoning** WHO poisoning page.
- Only **47% of WHO Member States had a poisons centre as of 1 January 2023** WHO poisoning page.

WHO links poison centres to **International Health Regulations (2005)** capacity for surveillance, detection, preparedness and response to chemical public-health events WHO departmental update, WHO IHR Benchmark 18.

1.2 Key Recommendations

WHO-based GRADE recommendation table

> **GRADE interpretation used:**
> **High** = further research unlikely to change confidence; **Moderate** = further research may change confidence; **Low** = further research likely to change confidence; **Very low** = estimate uncertain.
> **Strong recommendation** = desirable consequences clearly outweigh undesirable consequences for most settings.
> **Conditional recommendation** = context, feasibility, resources or patient values may substantially influence implementation.

Recommendation	GRADE Evidence Quality	Strength
Establish or strengthen a national or regional poison centre that provides specialized advice on the **diagnosis and management of poisoning**.	Moderate	Strong
Ensure poison centres support **24/7 or rapid-access clinical toxicology advice**, risk assessment, triage, referral and follow-up where feasible.	Moderate	Strong
Integrate poison centres into national **chemical-event preparedness and International Health Regulations** surveillance and response systems.	Moderate	Strong
Develop poison-centre functions beyond telephone advice, including **toxicovigilance, prevention, training, data collection, public outreach and chemical-incident response**.	Moderate	Strong
Build access to **clinical toxicology services**, including transfer pathways for severe poisoned patients where specialized care is needed.	Low	Strong
Build **analytical toxicology capacity** where feasible, especially for high-risk exposures, outbreaks, chemical incidents and unclear clinical syndromes.	Low	Conditional
Maintain structured access to evaluated toxicology databases, peer-reviewed information resources and harmonized poisoning data systems.	Moderate	Strong
Ensure availability and distribution systems for essential **antidotes, antivenoms and other substances used in poisoning**, including activated charcoal and specific antidotes where indicated.	Low	Strong
Use structured poison-call documentation: patient age, sex, weight, comorbidities, medications, pregnancy/lactation status, substance, amount, route, timing, symptoms and circumstances.	Low	Strong

Recommendation	GRADE Evidence Quality	Strength
Prioritize emergency stabilization before detailed history-taking: airway, breathing, circulation, seizures, coma, shock and exposure-source control.	Moderate	Strong
Use decontamination selectively: skin washing/eye irrigation for dermal/ocular exposures; activated charcoal only for appropriate recent ingestions when airway is protected.	Low	Conditional
Avoid inducing vomiting; ipecacuanha is discouraged because it may cause repeated vomiting, drowsiness and diagnostic confusion.	Low	Strong
For paracetamol/acetaminophen poisoning, use acetylcysteine when indicated by dose, timing and serum concentration protocols.	Moderate	Strong
For organophosphate/carbamate pesticide poisoning, prioritize airway support, decontamination and antidotal therapy; atropine is central, while oxime/pralidoxime use depends on compound, timing and local protocol.	Low	Strong for atropine/supportive care; Conditional for oximes
Use poisoning data for prevention policy, including pesticide hazard control and suicide-prevention strategies.	Moderate	Strong

Supporting rationale

- WHO defines poison centres as specialized services advising on **diagnosis and management of poisoning** and collecting data for prevention WHO poisoning page.
- The 2021 WHO guideline covers poison information services, clinical toxicology, analytical toxicology, toxicovigilance, chemical-incident preparedness, antidote availability, databases, training, quality management and funding WHO departmental update, RSU-hosted WHO PDF.
- WHO's INTOX training materials emphasize emergency stabilization, structured history-taking and documentation Trainer manual, Trainee manual.

- WHO pediatric hospital-care materials include poisoning sections covering ingested, dermal, ocular and inhaled poisons; activated charcoal; organophosphorus and carbamate compounds; paracetamol; salicylates; iron; opiates; and carbon monoxide WHO Hospital care for children.

1.3 Guideline Development Methodology

Evidence review process

The WHO poisoning-related guidance base is primarily composed of **technical guidance, operational standards, training manuals and public-health guidance**, rather than a single GRADE-formatted clinical treatment guideline for all poisonings.

Key methodological features identified:

- The 2021 WHO guideline updated the 1997 WHO/IPCS *Guidelines for poison control* because of:
  - renewed emphasis on poison centres under the **International Health Regulations (2005)**;
  - advances in technology;
  - changes in poison-centre operations since 1997 WHO departmental update.
- WHO's poisoning work includes guidance, training materials, global poison-centre directory support, harmonized data tools, pesticide exposure documentation tools and networking between poison centres WHO poisoning page.
- Canadian toxicovigilance experts contributed to the 2021 WHO poison-centre guideline, especially Chapter 5 on toxicovigilance and prevention Pan-Canadian Poison Centres 2021 Annual Report.

Consensus method

The retrieved summaries did not provide a full WHO guideline-development handbook description for the poisoning documents. However, the documents reflect consensus technical guidance from WHO/IPCS, poison-centre experts, clinical toxicologists and public-health stakeholders. WHO's broader meaningful-youth-engagement publication emphasizes that WHO guideline development should incorporate **evidence, equity and stakeholder values**, including youth voices where relevant WHO publication page.

Conflict of interest management

The supplied summaries did not identify detailed conflict-of-interest declarations for the five poisoning-related WHO documents. For future WHO poisoning-guideline updates, standard WHO guideline methods should include declaration, assessment and management of financial and non-financial conflicts, balanced guideline panels and transparent reporting.

II. Evidence Summary After WHO Latest Guideline Release (Case study: implementation of national guideline on the establishment of poison information control) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search strategy

Evidence was synthesized for the period after release of the latest WHO poisoning-specific guideline, **14 January 2021 to 29 April 2026**. The evidence scan prioritized:

- WHO official guidance and technical pages;
- poison-centre implementation studies and annual reports;
- national and regional poison-centre network reports;
- economic evaluations;
- clinical toxicology evidence summaries;
- systematic reviews, meta-analyses and GRADE-assessed reviews where available;
- clinical trial or trial-like evidence where identified.

Databases and sources searched

The supplied evidence summaries included searches of:

- WHO and WHO IRIS resources;
- PAHO/WHO;
- WHO International Health Regulations benchmark resources;
- national poison-centre networks including UK NPIS, Canada and United States;
- peer-reviewed implementation literature;
- RAND health-economic research;
- clinical toxicology evidence summaries;
- systematic review indexes and evidence aggregators.

Representative sources included WHO, PAHO/WHO, NPIS, Pan-Canadian Poison Centres, America's Poison Centers, RAND, *Global Health: Science and Practice*, NCBI Bookshelf, Science.gov-indexed evidence and toxicology education resources WHO poisoning page, NPIS 2024–2025 report, Global Health: Science and Practice, RAND.

Inclusion criteria

Included evidence met one or more of the following:

- addressed poisoning diagnosis, treatment, management or prevention;
- evaluated or described poison-centre establishment, function, quality, economic value or outcomes;
- provided WHO-aligned toxicovigilance, antidote, pesticide poisoning or chemical-event preparedness guidance;
- was published or operationally relevant between 2021 and 29 April 2026.

Exclusion criteria

Excluded or downgraded evidence included:

- non-poisoning WHO guideline-development publications, except where methods were relevant;
- sources without extractable poisoning diagnosis, treatment, management or implementation findings;
- preprints or non-peer-reviewed sources when higher-quality evidence was available;
- antidote recommendations not supported by the supplied source text.

2.2 Key New Studies

Evidence table with GRADE ratings

Study / source	Design	Key findings	GRADE Quality
Nepal Poison Information Center implementation report, 2024	Peer-reviewed implementation case study	Nepal's first institution-based Poison Information Center was established through a multilateral partnership; six months after inauguration it had supported nearly **250 cases** and showed early feasibility, with	Low

Study / source	Design	Key findings	GRADE Quality
		sustainability dependent on government support and service expansion Global Health: Science and Practice.	
Pan-Canadian Poison Centres 2021 Annual Report	National poison-centre surveillance and service report	Canadian poison centres managed **59%** of cases at home when benign outcomes were expected; among known outcomes, 50% had minor effects, 24% moderate, 5% major, 20% no effects and 0.4% died Pan-Canadian Poison Centres 2021 Annual Report.	Low
UK National Poisons Information Service 2024–2025 report	National service report	NPIS continued 24-hour specialist toxicology advice; user satisfaction was very good/excellent for 97.0% of NPIS telephone respondents, 87.5% for TOXBASE and 100% for UKTIS telephone service NPIS 2024–2025 report.	Low
RAND 2026 U.S. Poison Center Network impact study	Economic evaluation / benefit-cost analysis	U.S. poison centres generated estimated major societal value through reduced medical utilization, shorter hospital stays, mortality-risk reduction and public-health capacity; America's Poison Centers summarized estimated $3.1billion * *annualbenefitsand * *$**16.77** return per $1 spent RAND, America's Poison Centers.	Low to Moderate
U.S. National Poison Data System	National surveillance reports	Annual NPDS reports support toxicovigilance, case follow-up and	Low

Study / source	Design	Key findings	GRADE Quality
annual reporting, 2021–2024		national poisoning trend detection America’s Poison Centers.	
India Poison Information Centre, Karamsad, 2026	Implementation report / institutional news	New Poison Information Centre planned to provide **24/7 emergency toxicology advice and clinical support** to health professionals and the public NIHR RIGHT4 news.	Very Low
KidsToxLK Sri Lanka paediatric poison information and research centre, 2025	Implementation report / institutional news	Sri Lanka’s first dedicated paediatric poison information and research centre aimed to provide rapid evidence-based advice, workforce training and research into childhood poisoning NIHR RIGHT4 news.	Very Low
GRADE-assessed evidence base for organophosphorus pesticide self-poisoning	Systematic-review evidence summary	Identified **22 systematic reviews, RCTs or observational studies** assessing interventions including activated charcoal, atropine, benzodiazepines, gastric lavage, magnesium sulfate, oximes and others, with GRADE evaluation Science.gov.	Low to Moderate, depending on intervention
N-acetylcysteine adjunct in organophosphorus poisoning	Small randomized study summarized in evidence index	Thirty randomized patients; NAC was associated with biochemical antioxidant effects and lower atropine requirements, with no major adverse effects reported, but sample size was small Science.gov.	Low
Organophosphate poisoning clinical evidence summaries	Clinical review / expert synthesis	Supportive care, airway management, atropine and decontamination remain core; oxime evidence remains inconsistent and controversial StatPearls/NCBI Bookshelf.	Low

2.3 Evidence Gaps and Future Research Needs

Major gaps

1. **No single up-to-date WHO GRADE clinical guideline for all poisonings**
   - WHO provides strong technical and operational guidance for poison centres, but the supplied sources did not identify a current comprehensive WHO GRADE guideline covering all poisoning diagnosis, decontamination, antidotes and enhanced elimination.
2. **Limited high-quality trial evidence for many poisoning interventions**
   - Many interventions are supported by toxicological plausibility, observational data, case series, expert consensus and ethical/practical necessity rather than large RCTs.
3. **Antidote evidence remains uneven**
   - Acetylcysteine for paracetamol poisoning is well-established, but evidence for oximes in organophosphate poisoning remains inconsistent StatPearls/NCBI Bookshelf.
4. **Implementation outcomes need stronger comparative evaluation**
   - Poison-centre implementation studies show feasibility and value, but more quasi-experimental and controlled before–after studies are needed in low- and middle-income countries.
5. **Global poison-centre coverage remains inadequate**
   - Only 47% of WHO Member States had a poisons centre as of 1 January 2023 WHO poisoning page.
6. **Youth and community engagement is underdeveloped in poisoning guidance**
   - WHO's youth-engagement case study relates to abortion care, not poisoning WHO publication page. Future poisoning guidelines should consider engagement with adolescents, caregivers, agricultural communities, people with lived experience of self-poisoning, occupational groups and rural communities.
7. **Clinical trial registry evidence was not identified in the supplied summaries**
   - No specific poisoning-related clinical trial registry findings were available in the provided evidence set.

Priority research questions

Research question	Preferred design	Rationale
Does establishing a poison centre reduce mortality, unnecessary emergency visits, hospital length of	Stepped-wedge, interrupted time-series or controlled before–after study	Randomization may be difficult, but health-system outcomes are measurable.

Research question	Preferred design	Rationale
stay and costs in low-resource settings?		
Which poison-centre models are most feasible: national centre, regional hub, virtual network or hybrid model?	Comparative implementation study	WHO encourages poison-centre establishment, but optimal models vary by country.
What antidote stock levels are required for common national poisoning risks?	Needs assessment plus simulation modeling	Antidote access is repeatedly identified as essential.
In organophosphate poisoning, which subgroups benefit from oximes?	Pragmatic RCT or individual-patient-data meta-analysis	Current evidence is inconsistent.
What is the effect of pesticide hazard regulation on suicide and unintentional poisoning?	Natural experiment / policy evaluation	WHO identifies pesticide self-poisoning as a major global contributor to suicide.
How should youth and community voices be incorporated into poisoning-prevention and self-poisoning guidelines?	Mixed-methods guideline-development research	WHO's youth-engagement framework provides a model, but poisoning-specific evidence is absent.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Establish and strengthen poison centres.	Reinforced by Nepal implementation feasibility, Canada home-management outcomes, UK NPIS service performance and RAND economic evaluation Global	**Strong recommendation:** Every country should establish or designate a poison-information service, progressing toward 24/7 national coverage through national centres, regional hubs or networked models.

Original Recommendation	New Evidence Impact	Updated Recommendation
	Health: Science and Practice, NPIS 2024–2025 report, RAND.	
Provide expert diagnosis and management advice.	Supported by poison-centre reports showing triage, follow-up, case management and high user satisfaction NPIS 2024–2025 report, Pan-Canadian Poison Centres 2021 Annual Report.	**Strong recommendation:** Poison centres should provide rapid clinical risk assessment, triage, treatment advice and follow-up for health professionals and the public.
Integrate poison centres into chemical-event preparedness.	Reinforced by WHO IHR Benchmark 18 emphasizing surveillance and response capacity for chemical events WHO IHR Benchmark 18.	**Strong recommendation:** Poison centres should be embedded in national chemical-event surveillance, alert, response and risk-communication systems.
Support toxicovigilance and harmonized data collection.	Reinforced by Canada, U.S. NPDS and WHO toxicovigilance emphasis America's Poison Centers, WHO poisoning page.	**Strong recommendation:** Poison centres should collect standardized exposure, clinical and outcome data and report trends to public-health authorities.
Ensure antidote availability.	Reinforced by WHO guidance and clinical evidence showing the importance of timely antidotes, though stock-management evidence remains limited WHO feature, WHO Selection and Use of Essential Medicines.	**Strong recommendation:** Countries should maintain essential antidote lists, stockpiles and distribution plans based on local poisoning epidemiology.
Use activated charcoal selectively.	WHO pediatric guidance supports activated charcoal for selected recent ingestions; evidence varies by toxin and timing WHO AFRO Hospital care for children PDF.	**Conditional recommendation:** Consider single-dose activated charcoal for appropriate ingestions presenting early, generally within 1 hour, if the airway is protected and contraindications are absent.

Original Recommendation	New Evidence Impact	Updated Recommendation
Treat paracetamol poisoning with acetylcysteine when indicated.	WHO essential medicines material identifies acetylcysteine as a specific antidote; poison-centre training includes NAC use guided by serum concentration WHO Selection and Use of Essential Medicines, Trainee manual.	**Strong recommendation:** Use acetylcysteine promptly for paracetamol poisoning according to validated nomogram or local protocol.
Manage organophosphate poisoning with supportive care and antidotes.	Post-2021 evidence continues to support airway care, decontamination and atropine; oxime benefit remains uncertain Science.gov, StatPearls/NCBI Bookshelf.	**Strong recommendation for airway support, decontamination and atropine; conditional recommendation for pralidoxime/oximes** depending on compound, timing, severity and local toxicology guidance.
Build pediatric poisoning capacity.	Sri Lanka KidsToxLK and WHO pediatric hospital guidance reinforce the need for child-specific advice and prevention NIHR RIGHT4 news, WHO Hospital care for children.	**Strong recommendation:** Poison-centre systems should include pediatric dosing, child-specific risk assessment and childhood poisoning prevention.
Incorporate stakeholder engagement into guideline development.	WHO's 2025 youth-engagement case study is not poisoning-specific but provides a transferable model for inclusive guideline development WHO publication page.	**Conditional recommendation:** Future poisoning guidelines should include affected communities, youth, caregivers, agricultural workers, emergency clinicians and poison-centre users in guideline development.

Practical WHO-Based Clinical Management Algorithm for Poisoning

Immediate response

1. **Stabilize first**
   - Airway, breathing, circulation.
   - Treat seizures, coma, shock and severe respiratory depression.
   - Remove patient from exposure source.
   - Call a poison centre where available.
2. **Structured assessment**
   - Patient: age, sex, weight, pregnancy/lactation, comorbidities, medications.
   - Exposure: substance, formulation, amount, route, time, intent, co-exposures.
   - Clinical status: vital signs, toxidrome, mental status, pupils, secretions, skin findings, respiratory status.
   - Product information: packaging, label, concentration, pesticide/solvent/caustic status.
3. **Decontamination**
   - Remove contaminated clothing.
   - Wash skin; irrigate eyes.
   - Activated charcoal only for selected recent ingestions with protected airway.
   - Do **not** induce vomiting.
4. **Supportive and antidotal care**
   - Supportive care is universal.
   - Antidotes should be toxin-specific and protocol-based:
     - acetylcysteine for indicated paracetamol poisoning;
     - atropine for cholinergic organophosphate/carbamate poisoning;
     - oximes/pralidoxime selectively;
     - naloxone for opioid poisoning where clinically indicated, based on local protocols.
5. **Disposition**
   - Home observation only after poison-centre risk assessment and benign expected outcome.
   - Emergency referral for severe symptoms, high-risk substances, intentional self-harm, children with significant exposures, pregnancy, delayed-toxicity agents or uncertain exposure.
   - Follow up until outcome is known for moderate/severe cases.
6. **Public-health action**
   - Record the case.
   - Report clusters, unusual agents, product hazards and chemical incidents.
   - Use data to guide prevention, regulation and antidote planning.

Final WHO-Based Guideline Position as of 29 April 2026

1. The latest poisoning-specific WHO guideline identified remains **“Guidelines for establishing a poison centre”** from **14 January 2021** WHO departmental update.
2. WHO’s poisoning strategy is best understood as a **health-system and public-health infrastructure guideline**, centered on poison centres, toxicovigilance, clinical toxicology, analytical toxicology, antidote availability and chemical-event preparedness WHO poisoning page.
3. New evidence through **29 April 2026** strengthens the case for poison centres as clinically useful, economically valuable and essential for surveillance, but most implementation evidence remains **low to moderate quality** because it is observational or model-based.
4. Clinical poisoning care should remain anchored in **rapid stabilization, poison-centre consultation, structured exposure assessment, selective decontamination, supportive care, toxin-specific antidotes and toxicovigilance reporting**.
5. WHO has published a meaningful-youth-engagement case study for the **WHO Abortion care guideline (2022)**, not poisoning; future poisoning-guideline updates should adapt these inclusivity methods where relevant WHO publication page.